Utilizing Isolation Forest and XGBoost Algorithms in Forecasting Unusual Trip Behavior as a Commuter Safety Indicator

CHAPTER 1

Background of the Study

Public transportation connects people to schools, workplaces, and other daily needs. Personal safety remains a serious concern, especially for commuters who travel alone or who cannot easily call for help. These worries often depend on the time of day. A survey by UN Women (2021) in Kampala, Uganda, found that more than 80% of women felt unsafe on public transport late at night, compared to less than 8% during the daytime. While these findings come from another country, they show that safety concerns change across the day and deserve close attention alongside travel convenience.

In the Philippines, Corpuz et al. (2023) studied harassment on public transportation in Pampanga and found that young adults, particularly women aged 18 to 25, were most vulnerable to harassment and that reporting rates were low. Because many incidents go unrecorded, official records do not show the full extent of commuter danger. This creates a clear need for automated tools that can notice when a trip might be in trouble without waiting for a formal report, including in provincial areas such as Ilocos Norte.

The lack of reports creates a practical problem for computer-based safety tools. Since records of past incidents are scarce, a system cannot easily be trained on examples of dangerous trips. Waiting for a passenger to file a report also means that help comes only after an incident has occurred. An automated system must instead look at how the vehicle moves while the trip is happening. Some safety problems change the vehicle's movement, such as stopping in an unexpected place, leaving the normal route, or moving unusually slowly. Other problems leave no trace in movement, and a commuter who is frightened or unable to reach their phone cannot

always ask for help. A system that tracks movement cannot replace a commuter's own request for help, but it can provide an extra warning signal when something seems wrong.

Current safety tools often depend on a person to take action. Agapito et al. (2025) described Pathfinders, a Philippine commuter safety application with GPS tracking and an emergency button. However, this system still requires the user to press the button or needs a family member to watch a live map. Raw map points alone do not show whether a stop or a route change is normal or alarming. Some commercial ride-hailing companies already check trips automatically. Features like Uber's RideCheck (Kansal, 2020), Lyft's Smart Trip Check-In (Lyft, n.d.), and Grab's Safety Call (Sasikumar, 2026) monitor trips for unusual activity such as unexpected long stops or route changes. These services show that GPS and phone sensor data can highlight potential safety concerns during a trip, and they are the inspiration for the model in this study.

In academic research, Edwards and Hosein (2025) developed SayfTrip, which detects route deviations by checking how far a vehicle moves away from its recorded path using fixed distance boundaries. While helpful, this method has two clear limits. First, a vehicle can encounter trouble without leaving the road, such as a long, unexpected stop that stays within normal route boundaries. Second, fixed rules treat every road and hour the same way, even though a stop at a busy terminal at noon is normal, while the same stop on a quiet rural road at midnight is not. In addition, basic smartphone GPS has natural limits. Trees, tall buildings, and bridges can block satellite signals and cause location errors, making a normal trip look like an accidental detour (GPS.gov, n.d.). If a detector is too sensitive, it causes frequent false alarms that teach people to ignore warnings, but if it is too strict, it misses real problems (Edwards & Hosein, 2025).

Other researchers have studied unusual stops in long-distance coaches using GPS records (Deng et al., 2024; Sabir et al., 2025), and Li et al. (2021) found that driving events like long stops were linked to road safety risks. However, these studies look at events after they happen rather than predicting what will happen next, and they focus on coach operations and road

crashes rather than the safety of commuters on board. This leaves a gap in the literature. In the published work reviewed for this study, none provides a system that learns what is normal for each commuter from GPS data alone, measures how far a trip departs from normal habits, and forecasts upcoming risk states over a short forecast window without needing past incident records.

This study addresses that gap by building and testing a machine learning model pipeline that uses only basic phone GPS data: timestamp, latitude, and longitude. A prototype application records the trips and displays the model's output as a proof of concept, but this study focuses on the machine learning framework needed to interpret trip movements. It does not evaluate the application itself, including its usability, battery use, or the delivery of alerts. The proposed framework works in two connected stages. First, an Isolation Forest model (Liu et al., 2008) learns, for each commuter, the normal movement patterns on their routes, directions, and times of day without needing labeled incident data. It scores how unusual each moment is by looking at route distance and unexpected stops. Second, an XGBoost model (Chen & Guestrin, 2016) uses recent movement patterns to forecast the trip's concern level over a short time ahead, called the forecast window, whose length is tested in the study, reporting a confidence score for that prediction.

The proposed model treats unusual movement as an indicator that a trip may need attention, not as proof of a crime. It monitors changes in how a vehicle moves and does not claim to detect private interactions inside the vehicle or replace human judgment. The study evaluates the model on recorded field trips along local transit routes in Ilocos Norte, judging each trip with only that commuter's earlier trips, so the trip being judged was not part of the training data. Through these tests, the study measures how often the model raises false alarms during ordinary trips and evaluates how well it forecasts unusual movement patterns. This work examines how well tree-based machine learning methods can identify unusual travel behavior from basic trajectory records. The resulting model provides future developers with an empirical baseline and a technical starting point for building practical safety tools for everyday commuters.

Statement of the Problem

Safety concerns on public transport often go unreported, so commuters who travel alone or cannot call for help have few tools that notice, while a trip is under way, that it may need attention. This study therefore asks three questions.

1. How can GPS trips along local transit routes in Ilocos Norte be recorded every 10 seconds and cleaned, and how can they be turned into measures of speed, stopping, and distance from the route?

2. How can an Isolation Forest and XGBoost be used together to score how unusual each moment of a trip is, give a concern level from Normal to Critical, and forecast that level over a short forecast window?

3. How accurate are the forecasts compared with a simple baseline, how far can the probabilities be trusted, how many noted unusual events does the Model flag, and how many false alarms does it raise as more trips are added?

Objectives of the Study

General objective: To develop and evaluate a machine learning model that uses Isolation Forest and XGBoost to forecast unusual trip behavior from GPS records, as an indicator that a commuter's trip may need attention.

Specific objectives:

1. To record GPS trips along local transit routes in Ilocos Norte every 10 seconds using a prototype recording application, and to clean the records and measure speed, stopping, and distance from the route.

2. To build the Model by using an Isolation Forest to score how unusual each moment is, combining that score with route distance and unexpected stopping into a concern level from Normal to Critical, and using XGBoost to forecast the concern level over a short forecast window, whose length is compared in the study.

3. To evaluate the Model by testing each trip using only the trips recorded before it by the same commuter, and measuring how accurate its forecasts are against a simple baseline, how far its probabilities can be trusted, how many noted unusual events it flags, and how many false alarms it raises, as more trips are added.

Significance of the Study

This study examines whether basic GPS data and machine learning can help detect unusual travel behavior without needing incident reports or manual calls for help. Because this research focuses on building and evaluating the machine learning pipeline rather than releasing a completed mobile application, its value lies in testing the method and measuring how dependable it is. The findings may benefit the following groups.

Software developers and engineers. Creating safety tools is difficult when developers do not have records of past transit incidents or clear benchmarks for normal vehicle movement. This study provides developers with a technical starting point. It demonstrates how to clean raw GPS coordinates, turn them into useful measurements like speed and stopping duration, and combine an Isolation Forest with an XGBoost forecasting model. Developers can use this pipeline as the computing engine behind future mobile safety applications and automated check-in systems.

Future commuters. While commuters will not directly use an application during this study, they may benefit from the technical groundwork it creates. A safety tool is not helpful if it raises false alarms over routine traffic slowdowns, passenger pickups, or minor detours. By testing the model on real transit trips and measuring how often it raises false alarms, this study shows how dependable such a method is before it is built into tools for commuters.

Families and guardians. Parents and trusted contacts often worry about loved ones traveling alone, but they cannot monitor raw map coordinates continuously or tell whether an unexpected stop is harmless or concerning. This study examines whether a model can separate routine stops from unusual movement. If it can, future tools could give families alerts based on measured movement patterns rather than guesswork.

Researchers and the academic community. The tools and studies reviewed for this work mostly concern commercial ride-hailing services, coach operations, or city bus fleets. This study adds to academic research by showing how tree-based machine learning algorithms can detect and forecast unusual trip behavior using only basic smartphone GPS coordinates on everyday public transit routes. Other researchers and students can use this study's methods, feature designs, and baseline results to build better forecasting models and explore passive safety methods for public transportation.

CHAPTER 2
REVIEW OF RELATED LITERATURE

2.1 Introduction

This chapter reviews related studies and tools. It first looks at commuter safety in the Philippines and at the tools that already exist. It then follows the three objectives of this study: gathering and preparing GPS data, building the model, and evaluating the model. It ends with a summary of the gap in the literature and the conceptual framework.

2.2 Commuter Safety in the Philippine Context

Safety worries change with the time of day. A survey by UN Women (2021) asked 2,548 women in Kampala, Uganda about public transport. More than 80% felt unsafe late at night, compared with less than 8% in the daytime. This is another country, but it shows why the time of travel matters.

In the Philippines, Corpuz et al. (2023) studied harassment on public transport in Pampanga. Young adults, especially women aged 18 to 25, were the most affected, and reporting rates were low. Many incidents are never reported, so a safety tool cannot depend on past reports.

Commuters use different public vehicles. In Metro Manila, Okamura et al. (2013) studied what jeepney riders look for beyond fare and travel time, and security was one of the things they asked about. In Ilocos Norte, Baluan et al. (2026) studied jeepney and tricycle drivers in Sarrat, so both vehicles serve the province. These studies describe the setting, but they do not look at monitoring trips for safety.

Most safety tools need a person to act. Agapito et al. (2025) described Pathfinders, an app for Metro Manila with GPS tracking and an emergency button. The commuter must press the button, or a parent must watch the map. Some ride-hailing services check trips on their own. Uber's RideCheck notices an unexpected long stop and asks if help is needed (Kansal, 2020). Lyft checks rides for unusual activity such as long stops (Lyft, n.d.). Grab's Safety Call phones passengers after a sudden route change, a long stop, or a trip that is not moving forward

(Sasikumar, 2026). These features inspired this study: watch the trip, and check on the commuter when something looks unusual.

2.3 Gathering GPS Trip Data

A GPS trip record is a list of positions, each saved with its time. Each record has three parts: the time, the latitude, and the longitude. Edwards and Hosein (2025) saved these same three parts on a phone for each trip on a route. This study does the same with a prototype recording app.

How often a position is saved decides what the data can show. A short event can be missed if it falls between two records. Newson and Krumm (2009) found that their map matching got worse as the GPS rate got lower. Studies of long-distance intercity buses have the same problem. Their GPS often saves a position only every 30 seconds to two minutes, which can hide short stops (Deng et al., 2024; Sabir et al., 2025). Deng et al. (2024) say a 30-second gap is too long to tell whether a bus has stopped.

This study saves one position every 10 seconds. This is at least three times finer than the bus GPS in those studies, and it is fine enough to see the stops this study cares about. Sabir et al. (2025) give pauses under 40 seconds as an example of brief stops. With a record every 10 seconds, a 40-second stop has about four records, enough to show that the vehicle stayed still. A 30-second gap could give only one record or none. Edwards and Hosein (2025) note that phones can check location as often as once per second, although this uses more battery. A 10-second interval uses less battery while still being much finer than bus GPS. Other smartphone studies also used 10 seconds. Bagheri et al. (2020) sampled every 10 seconds while travellers were moving, to describe the trip route accurately enough, and Mears et al. (2021) requested a location every 10 seconds and note that this is a common recording interval for GPS data.

GPS is never exact. GPS.gov (n.d.) explains that buildings, bridges, and trees make positions less accurate. A position can be off, and a trip can look like it left the route when it did not. The records therefore have to be cleaned.

Recording trips also involves personal data. Under the Data Privacy Act of 2012 (Republic Act No. 10173), consent must be freely given, specific, and informed, and Section 12 lists when personal information may be processed. People who record or ride on the trips must give informed consent, and the records must be handled with care.

2.4 Cleaning and Preparing GPS Records

2.4.1 Cleaning GPS Errors

Raw GPS records need cleaning first. Zheng (2015) puts preprocessing early in the process. Common steps are removing noise, finding stay points (places where a vehicle stayed for a while), and map matching. Noise often comes from weak signals, such as in areas with tall buildings.

The first cleaning step removes GPS errors. A common way is to check the speed between two records. If a record implies a speed that a vehicle cannot reach, the record is an error. Chen et al. (2016) call speed thresholding probably the most popular method, but they note that the limit is hard to choose without knowing how the object moves, and that it can miss small errors. Verma et al. (2024) cleaned smartphone GPS data by removing records with a segment speed above 50 m/s (180 km/h) and records with an acceleration outside the range of -10 to 10 m/s^2. They also removed records with a reported error radius above 50 m. This study uses the same speed and acceleration limits. A speed of 50 m/s is 180 km/h, far above the highest limit in the Land Transportation and Traffic Code, which is 80 km/h for passenger cars and motorcycles on open country roads (Land Transportation and Traffic Code, 1964, Sec. 35). If a record implies a move faster than 50 m/s from the last accepted record, or an acceleration outside the range, it is treated as a GPS error and removed. The next record is then checked against the same last accepted record. This study records only time, latitude, and longitude, so it cannot use the error radius rule. Records that are duplicated, out of time order, or missing values are also removed. Small errors can still pass these checks, so later steps compare the measures with earlier trips as well.

2.4.2 Measuring Stops, Route Distance, and Speed

Stops matter a lot in this study. Deng et al. (2024) estimated stops from sparse coach GPS and separated unusual stops from normal ones. Sabir et al. (2025) compared how long a stop lasted with the usual stay in that part of the route. Both compare a stop with what is normal for the place. Sabir et al. (2025) also set a low-speed limit of 5 km/h to decide that a vehicle has stopped, and this study uses the same limit.

Distance from the usual route is also measured. Edwards and Hosein (2025) used the shortest distance between the current position and the positions of earlier trips on the same route. Newson and Krumm (2009) used map matching to find the most likely road behind noisy positions.

Speed, acceleration, and stopping time come from the positions and times. Li et al. (2021) did this for buses. They defined hard acceleration, hard braking, and long stops from acceleration and stop time. Edwards and Hosein (2025) link every point to its trip, so a new position can be compared with earlier trips on the same route.

2.5 Modelling Unusual Trip Behavior

2.5.1 Signals of Unusual Movement

Two signals are common in studies of unusual trips: leaving the expected route, and stopping where a stop is not expected. Edwards and Hosein (2025) put the distance from earlier routes into four levels: normal up to 150 m, minor up to 500 m, major up to 1,000 m, and critical beyond 1,000 m. They set these limits in advance. Ride-hailing features look for long stops and route changes (Kansal, 2020; Lyft, n.d.; Sasikumar, 2026). In a bus fleet, long stops, hard acceleration, and hard braking were each linked to crashes involving pedestrians and cyclists near bus stops, and long stops had the strongest link (Li et al., 2021). Deng et al. (2024) looked for unusual coach stops, such as stops at places where passengers should not be picked up. Their concern was coach safety, not harassment, so it supports finding unusual stops from GPS but not

what a stop means for a commuter. In these studies, unusual movement is treated as a possible warning sign, not as proof that something is wrong.

SayfTrip (Edwards & Hosein, 2025) saves each trip on the phone and compares each new position with earlier trips on that route. In the proof of concept, its detector looks only at location, not at timing. A vehicle can stop for an unusually long time near a familiar road without crossing a distance limit, and the fixed limits treat every part of the road the same.

2.5.2 Unsupervised Detection and the Isolation Forest

Labelled dangerous trips are rare, so methods that learn what is normal without labels are a good fit. The Isolation Forest (Liu et al., 2008) finds unusual points by how easily they can be separated from the rest, and it is fast and light. Liu et al. found that 100 trees built on samples of 256 points were enough, and this study uses the same settings. Zhang et al. (2011) used this idea on taxi GPS traces in iBAT. It found more than 90% of unusual trips with a false alarm rate under 2%. Chen et al. (2013) extended it in iBOAT, which compares a trip with normal routes for the same time period and points out which part is unusual. In one month of data from 7,600 taxis, they found about 43,800 unusual trips and concluded that most were likely deliberate fraud by drivers. So what is flagged depends on the setting, and a flagged trip is not a sign of danger by itself. Hariri et al. (2021) showed that standard isolation forests can give uneven scores and proposed the Extended Isolation Forest to fix this.

What is normal depends on the place and the time. iBOAT compares a trip with history for the same time period, and Sabir et al. (2025) compare a stop with the usual stay in the same part of the route. Fixed rules do not do this. A stop at a busy terminal at noon is normal, but the same stop on a quiet road at midnight is not. These studies support scoring each moment against the history of the same place and time. SayfTrip builds its history from the user's own earlier runs of a saved route (Edwards & Hosein, 2025), and this study does the same, so each commuter has an independent history. That history should stay recent, because old trips may no longer show what is normal (see Section 2.6.3).

2.5.3 Ordered Concern Levels

A safety tool should say how serious a situation is, not only whether it is unusual. Edwards and Hosein (2025) do this with four distance levels. Frank and Hall (2001) show a simple way to predict ordered levels with ordinary classification methods. The problem is turned into yes-or-no questions, such as "is the level above Minor?", and the answers are combined. This allows ordered levels such as Normal, Minor, Major, and Critical. For the Isolation Forest score and the length of a stop, the levels need a rule for how far from normal is far enough. A robust score does this by measuring the distance from the median in units of the median absolute deviation, so a few extreme trips do not distort it. Leys et al. (2013) recommend 2.5 as a cut-off, and Iglewicz and Hoaglin recommend labelling scores above 3.5 as potential outliers (as cited in NIST/SEMATECH, n.d.). This study uses 2.5 for Minor and 3.5 for Critical, and puts Major halfway at 3.

2.5.4 Making Labeled Data from Unlabeled Trips

XGBoost is a supervised method, so unlabeled trips have to be turned into labeled data first. It learns from examples that come with answers, called labels. The trips in this study have no labels, because nobody marks which moments were unusual. Other studies made labels from the data when labels were scarce. Bakker et al. (2025) trained models on labels made from Isolation Forest and autoencoder scores, and Yuan et al. (2024) gave pseudo-labels to unlabeled samples.

In this study, the labels are made in three steps from what the earlier sections described. First, each moment of a cleaned trip gets its measures (Section 2.4.2) and a concern level from Normal to Critical, which combines the Isolation Forest score, the distance from the route, and the stopping time (Sections 2.5.2 and 2.5.3). Second, each moment gets its label, which is the worst concern level in the forecast window, the short time ahead that the Model looks at. This uses the records that come after the moment. Third, the measures at the moment come only from the present and the past. They are the inputs, and the label is the answer. The window starts at

one minute, because a forecast has to arrive in time for a check-in. In SayfTrip, when a route deviation reaches the critical level, the user has 30 seconds to answer before a message goes to an emergency contact (Edwards & Hosein, 2025), so a window shorter than that would leave no time for that step. No study reviewed gives the best length, so this study compares other lengths. Near the end of a trip, less than the forecast window remains, so the label uses the records that are left. This turns unlabeled trips into labeled data without anyone marking incidents. The label shows what looks unusual, not what is dangerous.

2.5.5 Forecasting with XGBoost

To forecast over the forecast window, we need a method that works well on tables of measures such as speed and stopping time. Chen and Guestrin (2016) introduced XGBoost, a fast tree boosting method that is highly effective and widely used. Djon et al. (2023) compared five methods for forecasting theft in Chicago using only place and time, and XGBoost did best, with an F1-score of 0.86. XGBoost learns from labeled examples, which is why the labels described in Section 2.5.4 are made first. The label in this study is not a recorded incident. It is a concern level made from the unusualness score and the distance and stopping rules. The forecast therefore shows what is likely to look unusual, not what is likely to be dangerous.

2.6 Evaluating the Model

Testing the Model has five parts, in the order of the third objective: how trips are tested, how the forecast is compared with a baseline and with noted events, how the Model behaves as its history grows and expires, how far the probabilities can be trusted, and how many false alarms it raises.

2.6.1 Testing Trip by Trip in Date Order

How a model is tested decides whether its results can be trusted. Roberts et al. (2017) showed that random testing makes errors look smaller when data are grouped by time, place, or group. They advised splitting the data along those groups. Trips come in order, so the model

should learn from earlier trips and be tested on later ones. One trip should never be split between learning and testing.

Gama et al. (2013) support prequential evaluation, where each new example is first used to test the model and then used to update it. Replaying the trips in date order, and judging each trip using only the earlier trips, follows the same idea.

2.6.2 Comparing with a Baseline and with Noted Events

A forecast is only useful if it does better than a simple guess. This study compares the forecast with a simple baseline, such as assuming the current level stays the same. This shows whether a gain is more than the tendency of unusual behavior to continue. Labels for unusual events are few. Sabir et al. (2025) got strong results with only 10 labelled stops. So it can also help to check the Model against the few unusual events that are noted during trips. Sørbø and Ruocco (2024) show that the metrics for detecting unusual events in time series differ widely and that the choice should follow the task, so this study reports its measures separately and says what each one counts.

2.6.3 Learning as History Grows and Expires

A model that learns from history does poorly at the start, when there is little history. Reiss et al. (2024) call this the cold-start problem. Detectors have no data when they start, and methods that use no data are often not accurate enough. So this study reports how accuracy, probability quality, and false alarms change as more trips are added. Silvey and Liu (2024) use learning curves, which follow performance as the amount of training data grows, to judge how much data a model such as XGBoost needs, and this study does the same with trip history.

History should not grow forever, though. Roads, routes, and traffic change, so very old trips may no longer show what is normal today. Gama et al. (2013) found that forgetting old data, either with a sliding window or with weights that fade, helps a learner follow change. The survey of Gama et al. (2014) describes the same two ways to forget. The trade-off is that a short memory follows change quickly but has less data to learn from. This study therefore lets trips expire after

a set period, and it compares different periods to see which gives the fewest false alarms without losing accuracy. Cawley and Talbot (2010) warn that settings chosen on the same data used to judge them can look better than they really are, so the chosen period should then be checked on new trips.

2.6.4 Trusting the Probabilities

The probabilities also need checking. Niculescu-Mizil and Caruana (2005) showed that boosted trees tend to give distorted probabilities, and that calibration methods, such as Platt scaling and isotonic regression, improve them. Guo et al. (2017) measure this with the expected calibration error, which is the gap between confidence and how often the model is right. Conformal prediction gives answer sets that contain the true answer at a chosen rate (Angelopoulos & Bates, 2021), and adaptive versions keep this rate when data change (Gibbs & Candès, 2021). The Model holds out the latest 20% of the earlier trips to calibrate the probabilities. Hyndman and Athanasopoulos (2021) say a test set is typically about 20% of the sample, and this study uses the same share as a guide.

2.6.5 Counting False Alarms

False alarms must be counted. Edwards and Hosein (2025) note that false alarms could make users ignore warnings. A review of alarm fatigue in hospitals describes it as people not responding because of too many alarms, and links it to high false alarm rates and weak positive predictive value (Cvach, 2012). That is a hospital setting, but it shows why a safety tool should report how often it raises an alarm on ordinary trips. Edwards and Hosein tested their detector with four controlled deviation scenarios on one route. This study checks it with notes made during real trips.

2.7 Synthesis and Research Gap

The studies reviewed support each step of this study, but they do not join the steps together. Safety worries change with the hour, and many incidents are not reported, so tools

cannot depend on incident records (UN Women, 2021; Corpuz et al., 2023). Existing tools either need a person to act or check trips for unusual stops and route changes. SayfTrip uses fixed distance limits and was tested with controlled scenarios on one route (Agapito et al., 2025; Edwards & Hosein, 2025; Kansal, 2020; Sasikumar, 2026). GPS studies show that how often positions are saved, and how accurate they are, limit what trip records can show (GPS.gov, n.d.; Newson & Krumm, 2009; Zheng, 2015). Isolation methods can find unusual routes and stops without labels, and XGBoost can forecast from tables of measures (Chen & Guestrin, 2016; Chen et al., 2013; Deng et al., 2024; Liu et al., 2008). Studies of testing say to test in time order, check the probabilities, and count false alarms (Cvach, 2012; Gama et al., 2013; Guo et al., 2017; Roberts et al., 2017).

In the published work reviewed for this study, none joins these pieces for commuter safety. None learns what is normal for a commuter on a route, in a direction, and at a time of day from GPS alone, scores how far a trip departs from it in location and stopping, forecasts over a short forecast window, keeps its history recent, and reports how reliable the forecast is and how many false alarms it causes as more trips are added. This study fills that gap through its three objectives.

2.8 Conceptual Framework

Figure 1 shows the conceptual framework. The input is the GPS records of each trip (time, latitude, and longitude), a route reference for each direction, and notes made during the trip. The process has two steps. First, GPS errors are removed, and the records are cleaned and turned into measures of speed, stopping, route progress, and distance from the route. Second, an Isolation Forest scores how unusual each moment is, and the score is combined with distance from the route and unexpected stopping to give a concern level from Normal to Critical. These concern levels are used to make labels, and XGBoost learns from them to forecast the concern level over the forecast window, with a calibrated probability and a measure of confidence. The output is the concern level, the forecast, and its probability and confidence, plus the evaluation

results: forecast accuracy against a simple baseline, how far the probabilities can be trusted, how many noted unusual events are flagged, and how many false alarms are raised. Each trip is judged using only that commuter's earlier trips and is then added to that commuter's history. Trips older than a set period expire, so what counts as normal stays current.

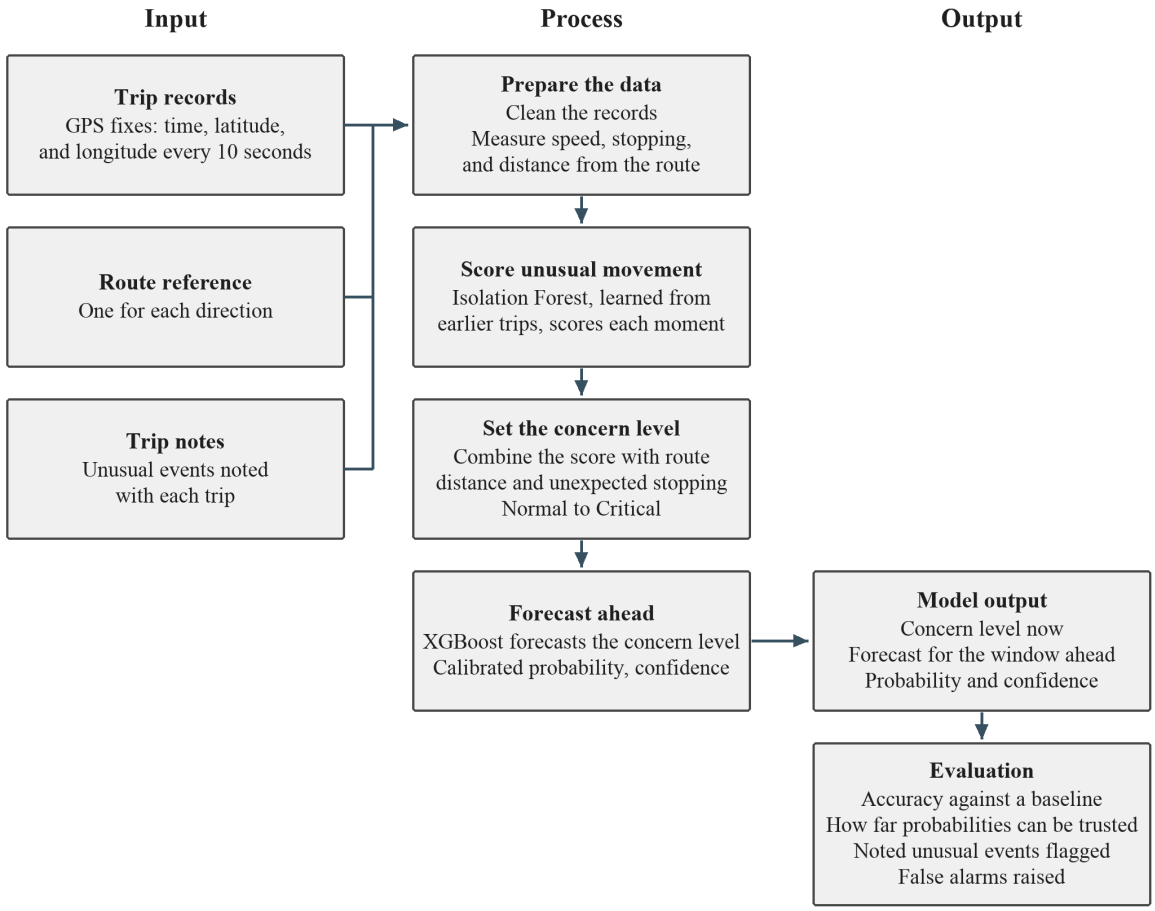


Figure 1. Conceptual framework of the study

CHAPTER 3
METHODOLOGY

3.1 Introduction

This chapter explains how the study was planned. It starts with the research design, which has three parts that match the three objectives. It then describes the study setting, the participants, and how the sample was chosen. The next three sections follow the objectives in order: data gathering and preprocessing (Objective 1), model development (Objective 2), and model evaluation (Objective 3). The chapter ends with the tools used and the ethical steps taken. Each setting that has a number is given with the reason for it. Table 3.1 at the end of the chapter lists them all in one place.

3.2 Research Design

This study uses a quantitative design with three parts, one for each objective. The first part is data gathering and preprocessing. Volunteer commuters record GPS trips with the prototype app, the records are cleaned, and they are turned into measures of speed, stopping, and distance from the route. This part belongs to the research design because the later parts can only be as good as the data they receive, and the choices made here, such as the recording interval and the cleaning limits, set what the Model can see. The second part is model development. The Model learns what is normal for each commuter from that commuter's earlier trips, scores how unusual each moment is, sets a concern level, makes labels from the unlabeled trips, and forecasts the level over a short forecast window. The third part is model evaluation. The recorded trips are replayed in the order they were recorded, as if each trip were happening for the first time, and the Model is judged on forecast accuracy, how far its probabilities can be trusted, the noted events it flags, and its false alarms. There is one Model, with the same settings for every commuter. What differs is the data it receives: each commuter has an independent history, so the Model learns from, and is judged on, that commuter's own trips. Each part uses what the one before it produced. Figure 2 shows the three parts and the steps in each.

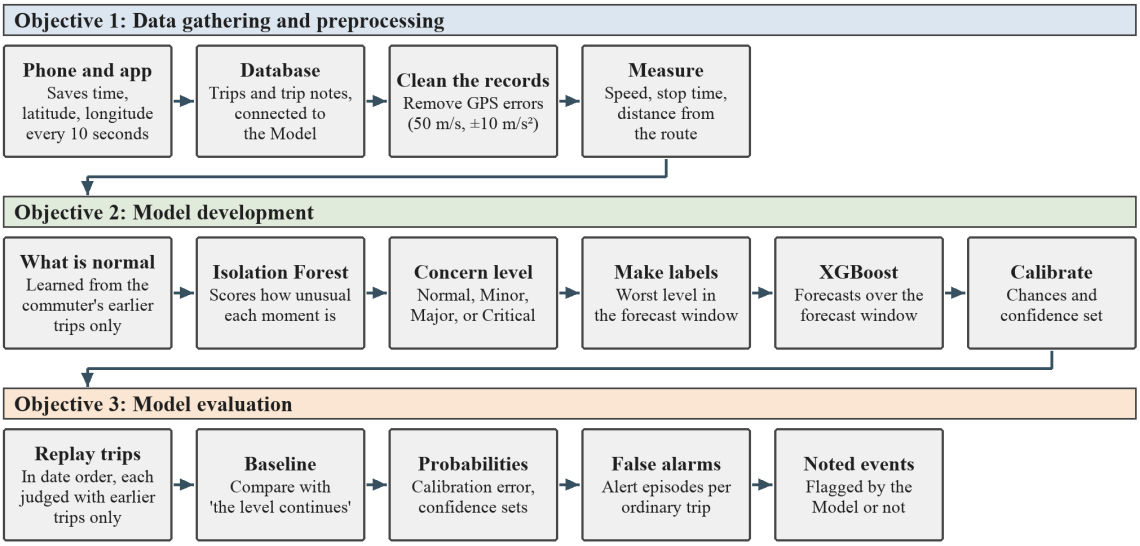


Figure 2. The three parts of the study and the steps in each.

The replay follows the advice of Roberts et al. (2017) and Hyndman and Athanasopoulos (2021), who say that data connected by time should be tested by learning from earlier records and judging on later ones. The study does not follow commuters after the trips, and it does not use incident reports. Safety incidents are scarce and many are never reported (Corpuz et al., 2023). The Model therefore learns what is normal from the trips themselves. Its output shows that a trip looks unusual. It does not show that anything harmful has happened.

3.3 Study Setting, Participants, and Sampling

3.3.1 Study Setting

The study takes place in Ilocos Norte. Trips are recorded on local transit routes in the province. The study is not limited to one kind of vehicle. Any public transport vehicle on a chosen route can be recorded, such as a jeepney, a tricycle, a bus, or a van, because the app saves only the position of the volunteer's phone and does not depend on the vehicle. Each volunteer notes the vehicle type for every trip, because vehicles can differ in how they travel. Some are faster than others, some stop more often or for longer to pick up passengers, and some may take slightly different roads on the same route. These differences change how long a trip takes and

what a normal stop looks like. Noting the vehicle type lets the results be checked for each type, and it shows whether a stop or a slow stretch that looks unusual is simply normal for that kind of vehicle. If trips from different vehicle types are mixed in one history, the Model learns one idea of normal for all of them. A stop that is unusual for a bus may look normal because a tricycle often stops there, so the Model may miss it. This is a limit of the study. Each volunteer rides their own usual route. A commuter's route has its own route reference, and each commuter has an independent trip history. The trips of one commuter never enter the history of another.

3.3.2 Participants

The participants are volunteer commuters. Each volunteer records trips on a route they already ride. To take part, a volunteer must be an adult, must ride one of the chosen routes regularly, must own a smartphone that can run the prototype recording app, and must give informed consent. Researchers do not record trips, so the data come from ordinary commuting and not from trips made only for the study.

3.3.3 Sampling Method

The study uses purposive sampling. In this method, participants are chosen because they have the characteristics the study needs (Etikan et al., 2016). Here the needed characteristic is regular riding on a chosen route. This method has a limit: because the sample is chosen on purpose, it may not represent all commuters in the province (Etikan et al., 2016). The results are therefore reported for the routes and volunteers studied.

3.3.4 Sample Size and Recording Period

The sample is counted in trips, because the Model learns from trips and not from people. Two things decide how many trips are needed.

The first is seeing an unusual event at least once. Viechtbauer et al. (2015) give a simple formula for the number of trips needed:

$$n = \frac{\ln(1-\gamma)}{\ln(1-\pi)} \tag{3.1}$$

where n is the number of trips needed, γ is the confidence that the event is seen at least once, and π is the chance that the event happens on one trip. With the example values of the authors, a 5% chance on one trip ($\pi = 0.05$) and 95% confidence ($\gamma = 0.95$), the working is:

$$n = \frac{\ln(1-0.95)}{\ln(1-0.05)} = \frac{\ln(0.05)}{\ln(0.95)} = \frac{-2.9957}{-0.0513} \approx 58.4 \tag{3.2}$$

Rounded up to a whole trip, about 59 trips are needed. Because each commuter has a personal history, these trips are needed from every volunteer. The study therefore records at least 60 trips from each volunteer, 30 in each direction. This keeps the two directions equal, and each direction has its own route reference and history. The first results use one volunteer with 60 trips recorded over about 30 days. More volunteers are added later, each with at least 60 trips, so that the study can show the Model works for more than one person and on more than one route. If unusual events turn out to be rarer than 5% of trips, the number of noted events will be small. In that case the study reports them as counts and does not claim a detection rate.

The second is having enough history to learn from. Silvey and Liu (2024) show that the amount of data a model needs can be judged by following performance as the amount of training data grows. The study does this too, and its smallest history is five earlier trips and its largest is the full set for one commuter's route direction. Having 30 trips per direction allows the largest history groups to be compared with the smallest.

Trips and days are used for different things in this study. The number of trips decides how much the Model has to learn from. The number of days decides how old that history is allowed to be. The two are linked by how often trips are recorded. The plan is about one trip in each direction on each recording day, which is two trips a day on a route. At that rate, 60 trips take about 30 recording days.

The recording period must be longer than the longest history length that is compared. The study compares keeping all earlier trips with keeping only the last 14 or 7 days (Section 3.6.4). In

the first 14 days, a 14-day window leaves nothing out, so it cannot be told apart from keeping all trips. A recording period of about 30 days therefore lets about half of the judged trips be judged with a window that really leaves out older trips. At the planned rate of a trip in each direction on each day, 30 days also gives the 60 trips needed. Volunteers record the trips they already make, so the study does not assign them to routes. A commuter's history then reflects that person's own usual vehicles, drivers, and times of day, which is what the Model should learn for that person.

More than two trips may be recorded on one day. They are all kept, and the history then holds more trips for the same number of days. A short window such as 7 days would then contain more trips than it would at two a day. For this reason the evaluation reports both: the number of earlier trips in the history for every judged trip (Section 3.6.3), and the number of days the history covers (Section 3.6.4).

3.4 Data Gathering and Preprocessing (Objective 1)

3.4.1 The Recording App

The prototype recording app saves three things for every record: the time, the latitude, and the longitude. These are the same three parts that Edwards and Hosein (2025) saved on a phone for each trip. The app is only a way to collect data. The study does not test the app itself, such as how easy it is to use or how much battery it uses.

3.4.2 Recording Interval

The app saves one position every 10 seconds. This is short enough to catch the stops this study cares about. Sabir et al. (2025) describe pauses under 40 seconds as brief, and a 40-second stop has about four records at 10 seconds. It is also much finer than the 30 to 60 seconds common in coach GPS, which Deng et al. (2024) say is too long to tell whether a vehicle has stopped. Bagheri et al. (2020) and Mears et al. (2021) also used 10 seconds for smartphone travel data. Edwards and Hosein (2025) note that polling every second uses more battery, which is a reason not to use the shortest interval.

3.4.3 Recording Procedure

A volunteer starts the app when boarding and stops it on arrival. The app is connected to a database, and the records of each trip are sent there. Each trip is stored with the volunteer's code, the route, the direction, the vehicle type, and the date. The database is connected to the Model, so the Model reads the trips of each commuter straight from the database and is fitted on them as that commuter's history grows. The trips of each commuter are kept together, with one set for each direction, as in the test data used to build the tools.

3.4.4 Route Reference

Each commuter's route has a route reference for each direction, which is the road line that vehicles usually follow from start to end. It is drawn on the map for the route and is used to measure how far along the route a vehicle is and how far it is from the usual road. Edwards and Hosein (2025) also compare positions with a saved route. Figure 3 shows the route reference of the first route. Point A is where the recorded trips start and point B is where they end.

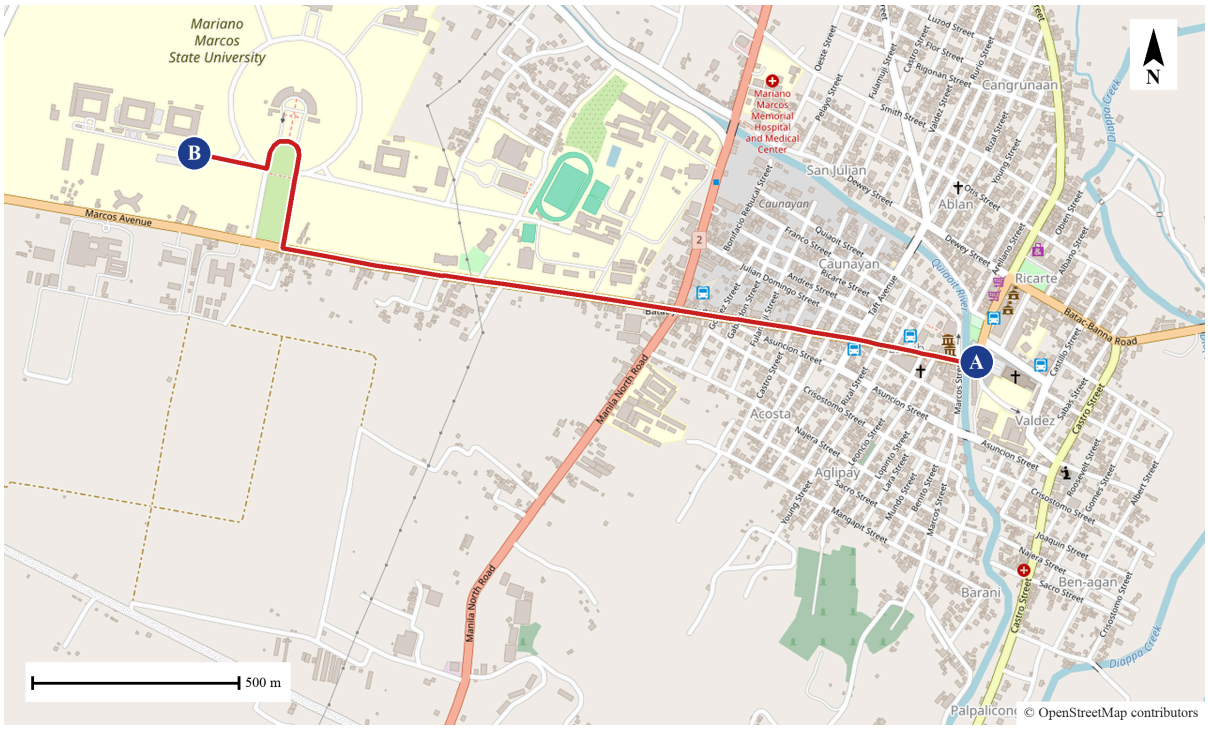


Figure 3. Route reference of the first route. Point A is the start of the recorded trips and point B is their end. Map data © OpenStreetMap contributors.

3.4.5 Trip Notes

The trip notes are written inside the app. After each trip, the app shows the volunteer a short form. The form asks whether anything unusual happened and, if so, what kind of event it was (a long unexpected stop, leaving the usual route, feeling unsafe, or another event), about when it happened, and a line of description. The app sends the notes to the database together with the trip, so the Model reads the notes with the trip they belong to. These notes are the only labels the study has, and they are kept apart from the model building. They are used only in the evaluation to check whether the Model flagged the noted events (Section 3.6.2). Sørbø and Ruocco (2024) show that the way unusual events are counted changes the results, so the notes ask for a time and the evaluation counts an event only inside a stated time window.

3.4.6 Cleaning GPS Errors

Each file is cleaned before anything else, because preprocessing comes early in the process (Zheng, 2015). First, rows with a missing or unreadable time, latitude, or longitude are removed, the rows are put in time order, and repeated times are removed. Second, if the file has an accuracy column, records with a reported error radius above 50 m are removed. Third, a record is removed if it implies a move faster than 50 m/s from the last kept record. Fourth, a record is removed if it implies an acceleration outside the range of -10 to 10 m/s². The speed, accuracy, and acceleration limits come from Verma et al. (2024). The speed and acceleration checks are repeated until no record breaks a limit. At a 10-second interval the acceleration limit is rarely reached, and the speed limit does most of the work. A speed of 50 m/s is 180 km/h, far above the highest limit of 80 km/h in the Land Transportation and Traffic Code (1964).

The study records how many records were removed by each rule for every trip. A trip is set aside if it loses more than one record in five or has fewer than 20 records left, because it can no longer be followed reliably. The list of trips set aside, and the reason for each, is reported with the results so that the rule can be reviewed.

3.4.7 Measuring Speed, Stops, and Distance from the Route

From the clean records, the following measures are worked out for every moment of a trip. Speed is the distance between two records divided by the actual time between them, which avoids errors when a record is late or missing. Acceleration and turning come from the changes in speed and direction. A vehicle counts as stopped when its speed is below 5 km/h, the low-speed limit used by Sabir et al. (2025). Stop time is how long it has stayed below this speed.

Every position is projected onto the route reference for its direction. This gives the route progress, which is how far along the route the vehicle is, and the distance from the route line. This is similar to the map matching of Newson and Krumm (2009). The study also measures the shortest distance between the position and the positions of earlier trips on the same route, as Edwards and Hosein (2025) did. Speed, stopping, and route distance are also summarised over

the minute before each moment, The look-back stays at one minute, even when the forecast window is changed. Figure 4 shows route progress and the distance from the route for one position.

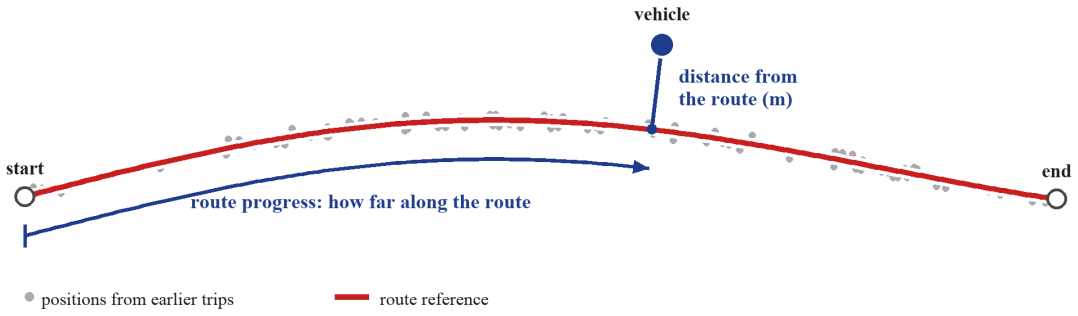


Figure 4. Route progress and distance from the route for one position of a vehicle (illustration).

3.5 Model Development (Objective 2)

3.5.1 What Is Normal for Each Place and Time

The Model first learns what is normal for a commuter from that commuter's own earlier trips. It divides each route direction into stretches of road. For each stretch, and for each part of the day, it learns the usual speed and how much speed varies. What is normal depends on the place and time, as in iBOAT (Chen et al., 2013) and in the stop comparison of Sabir et al. (2025). The day is split into four parts: night, morning, midday, and afternoon. Night is kept apart because UN Women (2021) found that safety worries are much higher at night. The split is also tested: Section 3.6.5 compares the four parts with a split into night and day only and with no split at all. Where a stretch has few trips in one part of the day, the Model leans on the all-day figure for that stretch. The first stretch length is 100 m, and the Model leans on the all-day figure with a strength of 8 trips. No source gives these values, so Section 3.6.5 tests other values around them.

3.5.2 Scoring Unusual Movement with the Isolation Forest

An Isolation Forest (Liu et al., 2008) gives each moment a score for how easy it is to separate from the rest. Liu et al. found that 100 trees built on samples of 256 points were enough, and this study uses the same settings. Two forests are built, each from half of the history trips. A history trip is scored by the forest that did not see it, and a new trip gets the average of both. This stops the Model from scoring trips it has already memorised, which would make the scores look too normal. Cawley and Talbot (2010) warn about the same problem of judging on data that was used to build the model. Scores are smoothed over 30 seconds so that a single odd record does not decide the level. Figure 5 shows how this works.

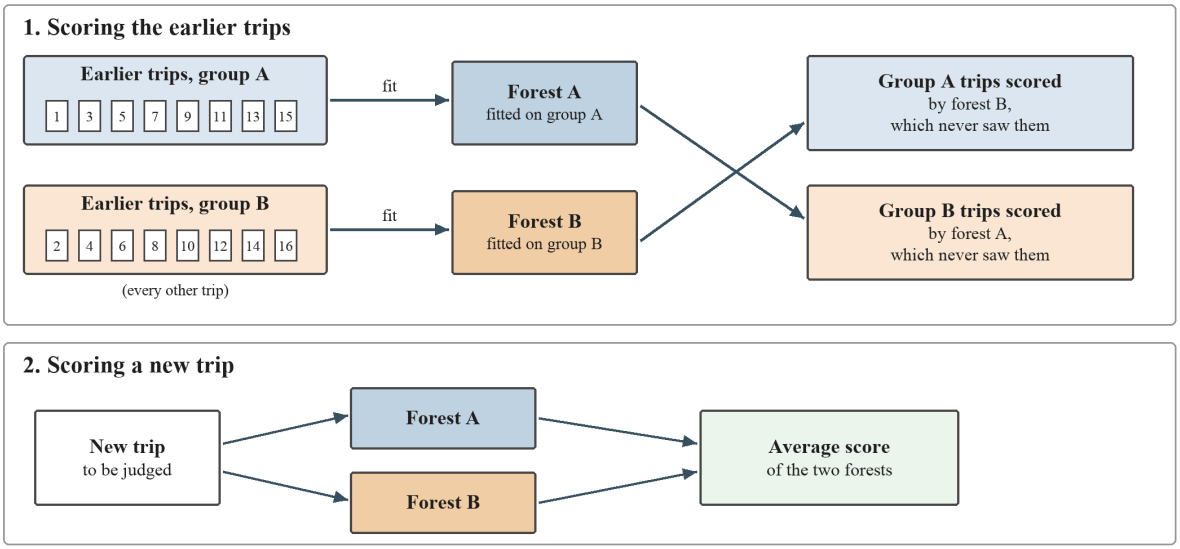


Figure 5. How the two Isolation Forests score the earlier trips and a new trip.

3.5.3 Setting the Concern Level

Each moment gets the highest of three levels. The first level comes from the forest score. The second comes from the distance from the route: up to 150 m is Normal, up to 500 m is Minor, up to 1,000 m is Major, and beyond 1,000 m is Critical (Edwards & Hosein, 2025). While the commuter has fewer than 5 earlier trips in that direction, the distance is measured from the route line. After that, it is measured from the commuter's earlier trips, so a detour that the

commuter makes often becomes familiar and counts as Normal. The number 5 has no source, so it is treated as an experiment in Section 3.6.5. The third level comes from how long the vehicle has been stopped.

The forest score and the stop time are turned into levels with a robust score. This measures the distance from the median in units of the median absolute deviation, so that a few extreme trips do not distort it. The forest score is compared with the scores of the history. The stop time is compared with the lengths of stops that finished in the history, so a stop is judged against what is usual for the route and not against a fixed number of seconds, as in Sabir et al. (2025). Leys et al. (2013) recommend 2.5 as a cut-off, and Iglewicz and Hoaglin recommend labelling scores above 3.5 as potential outliers (as cited in NIST/SEMATECH, n.d.). The levels Minor, Major, and Critical start at robust scores of 2.5, 3, and 3.5, so Major lies halfway between the two recommended values. Figure 6 shows how the three levels are combined, with an example.

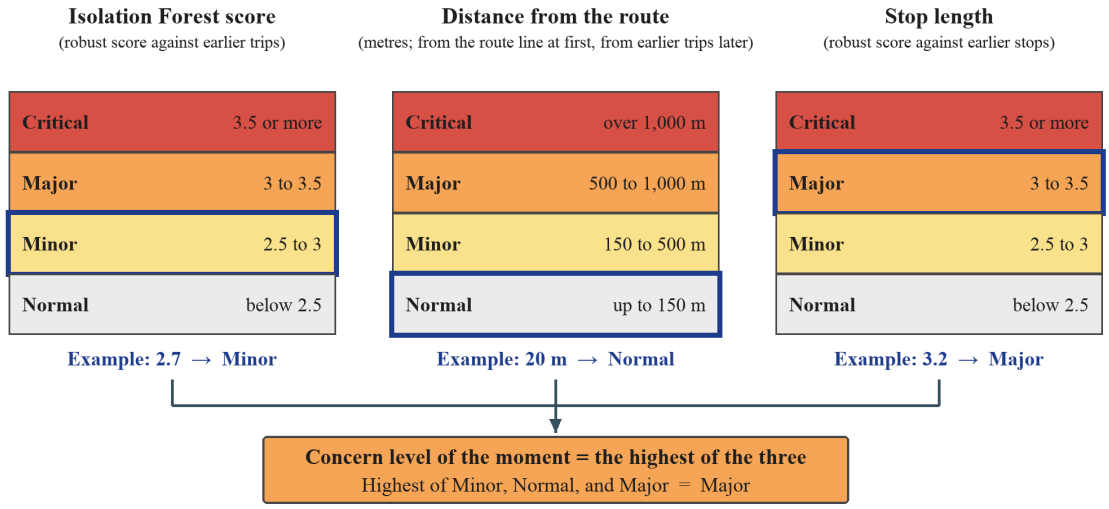


Figure 6. How the concern level is set from the forest score, the distance from the route, and the stop length (illustrative values).

3.5.4 Making Labelled Data from Unlabeled Trips

XGBoost learns from examples that come with an answer. The labels are made after the concern level is set, because they are built from it. The recorded trips have no answers, because nobody marks which moments were unusual. The study therefore makes the answers from the trips themselves, in four steps. Bakker et al. (2025) and Yuan et al. (2024) also made labels from unlabeled data to train supervised models.

First, every moment of a trip gets a concern level from Normal to Critical (Section 3.5.3). Second, for each moment, the Model looks at the forecast window, which is the short time that comes after it. Third, the label of the moment is the worst concern level found in that window. Fourth, the measures at the moment (Section 3.4.7), such as speed, stop time, and distance from the route, are the input, and the label is the answer that XGBoost learns to predict. Near the end of a trip, less than the forecast window remains, so the label uses the records that are left.

Figure 7 shows an example from one trip in the test data. The records on the left have only the time, the latitude, the longitude, and the Isolation Forest score. On the right, the speed, the stop time, and the distance from the route are added as inputs, and each moment has its label. For example, at 09:08:09 the vehicle is moving at an ordinary 28.9 km/h with no stop, but the label is Major because the vehicle stops soon after. In this way, XGBoost learns what the moments before a long stop tend to look like.

Unlabeled trip records

Timestamp	Latitude	Longitude	Isolation Forest score
09:07:49	18.05511	120.56098	0.43
09:07:59	18.05524	120.56021	0.42
09:08:09	18.05536	120.55946	0.38
09:08:19	18.05549	120.55854	0.38
09:08:29	18.05564	120.55756	0.39
09:08:39	18.05564	120.55751	0.44
09:08:49	18.05564	120.55751	0.48

Labelling →

Training data for XGBoost

Inputs					Answer
Time	Speed (km/h)	Stop time (s)	Route distance (m)	Isolation Forest score	Label
09:07:49	33.5	0	0.0	0.43	Normal
09:07:59	29.9	0	0.0	0.42	Normal
09:08:09	28.9	0	0.0	0.38	Major
09:08:19	35.6	0	0.0	0.38	Critical
09:08:29	37.5	0	0.0	0.39	Critical
09:08:39	1.9	10	0.1	0.44	Critical
09:08:49	0.0	20	0.1	0.48	Critical

Figure 7. Making labelled data from unlabeled trip records (example rows from one trip in the test data).

The forecast window starts at one minute, because a forecast has to arrive in time for a check-in. In SayfTrip, when a route deviation reaches the critical level, the user has 30 seconds to answer before a message goes to an emergency contact (Edwards & Hosein, 2025), so a window shorter than that would leave no time for that step. No study reviewed gives the best length, so the window is treated as an experiment and compared in Section 3.6.5. A longer window gives more warning, but it may make the forecast less accurate. The labels show what looks unusual compared with earlier trips. They do not show what was dangerous.

3.5.5 Forecasting with XGBoost

XGBoost (Chen & Guestrin, 2016) forecasts the worst concern level in the forecast window from the measures at each moment and the concern level now. It suits tables of measures such as speed and stopping time. The levels are ordered, so the problem is turned into three yes-or-no questions: is the worst level in the forecast window at least Minor, at least Major, and at least Critical? Frank and Hall (2001) show how to combine such answers into an ordered prediction. The three chances are kept in order, so a higher level never gets a higher chance than a lower one. The number of trees is chosen by early stopping on held-out trips. The other settings are kept small for a small data set, with shallow trees and a low learning rate, and are listed with

the code. Figure 8 shows the three questions and how the latest earlier trips are held out (Section 3.5.6).

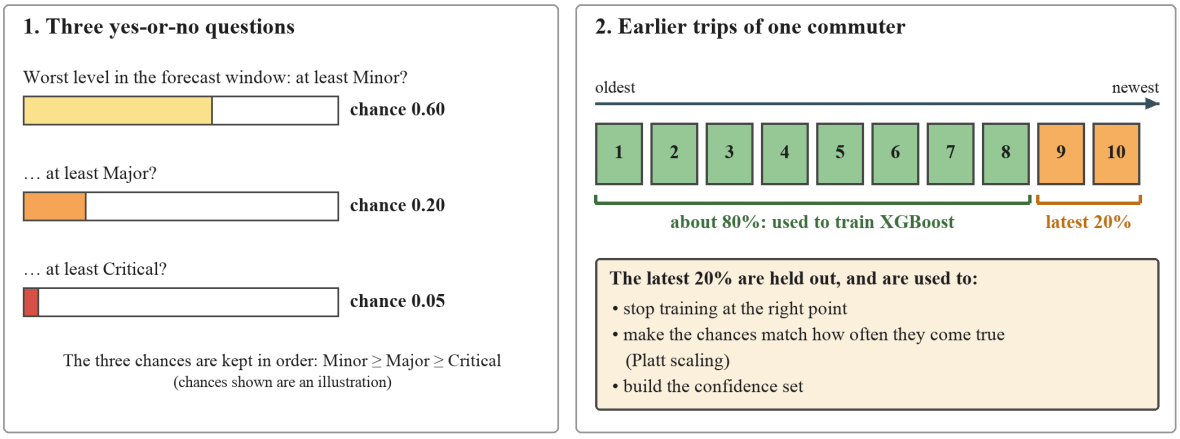


Figure 8. The three yes-or-no questions of XGBoost and the use of the latest 20% of earlier trips (illustrative chances).

3.5.6 Calibration and Confidence

The latest 20% of the history trips, and at least two trips, are held out when the Model is built. Hyndman and Athanasopoulos (2021) say a test set is typically about 20% of the sample, and this study uses the same share as a guide. These trips are used for early stopping, for calibration, and for the confidence sets. Boosted trees tend to give distorted probabilities, and Platt scaling improves them (Niculescu-Mizil & Caruana, 2005), so the chances from the three questions are calibrated with Platt scaling on the held-out trips. Guo et al. (2017) call the gap between confidence and how often a model is right the calibration error, and the study reports it.

The Model also gives a confidence set, which is a short list of levels that contains the true level at a chosen rate. This is split conformal prediction (Angelopoulos & Bates, 2021). The target rate is 90%, a common choice in conformal prediction. The cut-off is taken from the held-out trips.

3.5.7 Keeping the History Recent

Roads, routes, and traffic change, so very old trips may no longer show what is normal. Gama et al. (2013) found that forgetting old data helps a learner follow change, either with a window of recent examples or with weights that fade. The Model uses a window. It learns only from the same commuter's earlier trips that fall inside a chosen number of days before the trip being judged. The best length is not known in advance, so the study tests several lengths (Section 3.6.4).

3.6 Model Evaluation (Objective 3)

3.6.1 Testing Trip by Trip in Date Order

The trips are put in order by start time. Each trip is judged by the Model fitted only on the same commuter's trips before it, and never on another commuter's trips. After it is judged, the trip joins that commuter's history for the next one. A trip is never split between building and judging, because records from the same trip are closely related (Roberts et al., 2017). This is the prequential idea of Gama et al. (2013): each new example is first used to test the model and then to update it. A trip is judged only when at least five earlier trips are available. The study judges this way for the whole recording period. The results are worked out separately for each commuter and then summarised across commuters, with the lowest and highest values shown, so that no single commuter decides the average alone. Figure 9 shows which earlier trips are used in three cases.



Figure 9. Which earlier trips are used to judge a trip: early in the recording, keeping all earlier trips, and keeping the last 14 days (illustration; not every trip is shown).

3.6.2 Baseline and Noted Events

The forecast is compared with a simple baseline that assumes the current concern level continues. Both are scored with average precision, which rewards ranking real concerns above ordinary moments, for the questions "at least Minor" and "at least Major". The average gap between the forecast level and the true level is also reported. For each of the two questions, the study also reports the share of unusual moments that an alert caught and the share of alerts that were right. For "at least Minor", an alert from the forecast means a chance of at least 0.5 of Minor or worse, and an alert from the baseline means that the current level is Minor or higher.

The notes from the volunteers are used to check the noted events. A noted event counts as flagged if the Model raises an alert from two minutes before to two minutes after the noted time. The window allows for a volunteer's time estimate being rough. It is a starting value, so the study also reports one minute and three minutes. An alert is raised when the forecast chance of a level of Major or worse is at least 0.5, the plain midpoint. This is a starting value, and Section 3.6.5

compares lower limits. The study reports how many noted events were flagged by the forecast and by the current level, and how far ahead the first alert came. Without enough noted events, this check is reported as counts only.

The study also asks whether the forecast gives earlier warning than reacting to the current level. For each trip where both the forecast and the current level reach an alert, it measures how many seconds earlier the forecast alerted. It also counts the trips where both alerted, where only the forecast alerted, and where only the current level reached the alert level. These counts show how often the forecast agrees with the current level and how often it adds or misses something.

3.6.3 History Growth

To show how the Model behaves at the start, the judged trips are grouped by how many earlier trips were in the history, such as 5 to 9, 10 to 19, and so on. For each group, the study reports forecast accuracy against the baseline, calibration error, confidence-set coverage, and false alarms. This is a learning curve in the sense of Silvey and Liu (2024). It also shows the cold-start problem described by Reiss et al. (2024).

3.6.4 History Length

The study compares three history lengths on the same trips: all earlier trips, and only the last 14 or 7 days (see Figure 9 for the 14-day case). For each, it reports the same measures as above. The length with the fewest false alarms and no loss of accuracy is chosen. Cawley and Talbot (2010) warn that settings chosen on the same data used to judge them can look better than they are, so the chosen length is confirmed on new trips recorded after the choice was made.

3.6.5 Checking the Starting Values

Seven starting values have no direct source: the forecast window of one minute, the 100 m stretch length, the strength of 8 trips, the four parts of the day, the 5 earlier trips before the route distance switches from the route line to the earlier trips, the alert limit of 0.5, and the 20% of earlier trips held out for calibration. They are treated as experiments. The study repeats the

evaluation with forecast windows of 30 seconds and 2 minutes, with stretch lengths of 50 and 200 m, with strengths of 4 and 16 trips, and with the day split only into night and day or not split at all. For the route distance, it also compares switching after 3 and after 10 earlier trips, and the earlier rule that takes the larger of the distance from the route line and from the earlier trips. It also compares holding out 10% and 30% of the earlier trips, and alert limits from 0.1 to 0.4. The alert limit changes only the alerts and not the accuracy scores, so it is compared by the share of unusual moments caught, the share of alerts that were right, and the false alarms per hour. Changing the forecast window also changes the labels, so the results for different settings are compared by the gain, which is how far the forecast's average precision for Major or worse is above the baseline's, together with the calibration error, the false alarms per hour, and the lead over the current level. If the results change little, the starting values are kept. If they change a lot, the values are re-chosen on the earlier trips only and then confirmed on new trips.

3.6.6 Probabilities and Confidence

The study reports the calibration error for the chance of "at least Minor" and "at least Major". It uses equal-width bins of predicted chance. Guo et al. (2017) use 15 bins, and this study uses 8 so that each bin has enough records with a small data set. It also reports the share of moments for which the confidence set contained the true level, and the average size of the sets, and compares the share with the 90% target.

3.6.7 False Alarms

False alarms are counted on ordinary trips, which are the trips with no noted event. Alerts less than one minute apart are counted as one alert episode, which is the starting forecast window. The study reports the number of alert episodes per ordinary trip, the share of ordinary trips with at least one alert, and the number of alert episodes per hour of travel, for the forecast and for the baseline. This follows the concern of Edwards and Hosein (2025) and Cvach (2012) that too many false alarms teach people to ignore warnings.

3.6.8 Route Familiarization

To show that the Model learns a commuter's usual detours, the study takes the trips that go beyond 150 m of the route line and have at least 20 earlier trips in the same direction. It counts the moments beyond 150 m when the distance is measured from the route line, and when it is measured from the most recent 2, 5, 10 and 20 earlier trips. The fewer moments remain, the better the earlier trips describe the commuter's own paths.

3.6.9 A Fixed Distance Rule for Comparison

The forecast is also compared with a fixed rule that alerts whenever the vehicle is more than 150 m from the route line, which is the first level of the fixed bands in SayfTrip (Edwards & Hosein, 2025). The study counts the alert episodes per trip and per hour for both. It also counts how many long stops and how many episodes beyond 150 m each one flags. A long stop is a finished stop of 40 seconds or more, the length that Sabir et al. (2025) call brief. A rule flags a stop or an episode if it alerts from one minute before it starts to its end. Without trip notes, this comparison shows what each rule watches for, and not which alerts were right.

3.6.10 A Simpler Model for Comparison

XGBoost may be more than a small data set needs, so its forecast is compared with a plain logistic regression. This is a simple method that draws a straight boundary between moments that were followed by an unusual window and moments that were not. It uses a small set of the same inputs: the concern level now, the Isolation Forest score, the stop time and the excess stop time, the time off the route, the distance from the route, the speed, the speed trend, the time since the last stop, the stops in the last minute, and the speed score. It uses the same labels, the same calibration, and the same trips as XGBoost. The comparison asks whether the extra complexity of XGBoost is worth it for the amount of data available.

3.6.11 Other Checks

Three more checks make the results easier to read. The study shows which part of the concern level, the Isolation Forest score, the route distance or the stop length, set each level. It reports the results for each direction of the route. It also measures how long the Model takes to fit on the earlier trips and to score a trip, to see whether it is quick enough to be refitted after every trip.

3.7 Tools

The data preparation and the Model are written in Python. The Isolation Forest, the Platt scaling, and the evaluation measures use scikit-learn (Pedregosa et al., 2011). The forecasting uses the XGBoost library (Chen & Guestrin, 2016). The same code builds the Model and runs the replay, so the results can be repeated from the recorded files.

3.8 Ethical Considerations

Recording trips involves personal data. Under the Data Privacy Act of 2012 (Republic Act No. 10173), consent must be freely given, specific, and informed. Every volunteer is told what is recorded, why, who can see it, and how long it is kept, and signs a consent form before recording. Volunteers can stop at any time and ask for their records to be removed. Riders who share the vehicle are not recorded, because the app saves only the volunteer's own position. Files carry a volunteer code and not a name. The records are stored in the database, which only the researchers can open, and they are kept only as long as the study needs them. The study will ask for approval from a research ethics committee before any recording begins.

Start and end positions can show where a person lives or works. To lower this risk, results are reported for routes and not for people, and the maps in the report show only the public start and end places of a route and never a volunteer's home.

3.9 Summary of Settings

Table 3.1 lists each setting used in this chapter, its value, and where the value comes from. Settings marked as design choices have no source. They are chosen for simplicity, and the study reports them with the results so that they can be reviewed.

Table 3.1 Settings used in the study and their basis

Setting	Value	Basis
Recording interval	10 seconds	Sabir et al. (2025); Deng et al. (2024); Edwards & Hosein (2025); Bagheri et al. (2020); Mears et al. (2021)
Speed limit for cleaning	50 m/s	Verma et al. (2024); Land Transportation and Traffic Code (1964)
Acceleration limit	-10 to 10 m/s²	Verma et al. (2024)
Error radius limit	50 m	Verma et al. (2024)
Trip set aside	more than 1 record in 5 lost, or fewer than 20 records left	Design choice, reported with the results
Stopped speed	below 5 km/h	Sabir et al. (2025)
Route distance levels	150, 500, and 1,000 m	Edwards & Hosein (2025)
Level cut-offs for forest score and stop time	robust scores of 2.5, 3, and 3.5	Leys et al. (2013); NIST/SEMATECH (n.d.); Major is the midpoint
Isolation Forest size	100 trees, samples of 256	Liu et al. (2008)
Forecast window	starts at 1 minute; 30 seconds and 2 minutes are compared	Experiment. Edwards & Hosein (2025): a 30-second response time means a shorter window leaves no time (Section 3.6.5)
Held-out share of history	latest 20%, at least 2 trips; 10% and 30% are compared	Starting value after Hyndman & Athanasopoulos (2021); experiment (Section 3.6.5)
Confidence set rate	90%	Angelopoulos & Bates (2021) for the method; 90% is a common choice
Trips per volunteer	at least 60, 30 per direction	Viechtbauer et al. (2015): 59 trips at 5% and 95%
Recording period	about 30 days for each volunteer	So that about half of the judged trips come after day 14, when a 14-day window starts to leave out older trips; also gives 60 trips at two a day
History lengths compared	all, 14, and 7 days	Compared in the study (Section 3.6.4)
Road stretch and strength	100 m and 8 trips	No source; 50 and 200 m, and 4 and 16 trips, are tested (Section 3.6.5)
Earlier trips before the route distance uses them	5; 3 and 10, and the larger-of-both rule, are compared	Experiment (Section 3.6.5)

Parts of the day	night, morning, midday, afternoon; night and day only, and no split, are compared	Experiment (Section 3.6.5). Night is kept apart because of UN Women (2021)
Fewest earlier trips before judging	5	Design choice, reported with the results
Look-back for the measures	1 minute	Design choice; stays fixed when the forecast window changes
Alert rule	chance of Major or worse of at least 0.5; limits from 0.1 to 0.4 are compared	Plain midpoint as the starting value; experiment (Section 3.6.5)
Alert episode gap	60 seconds	Starting forecast window; design choice
Event window for noted events	2 minutes before to 2 minutes after; 1 and 3 minutes also reported	Experiment; reported with the results
Calibration bins	8	Guo et al. (2017) use 15; fewer bins suit a small data set

CHAPTER 4
INITIAL RESULTS AND DISCUSSION

4.1 Introduction

This chapter reports the first results of the study. They are called initial because they come from the trips of one commuter only, and because the trip notes have not been collected yet. The chapter follows the three objectives. It first describes the data and how they were prepared (Objective 1). It then shows what the Model produced (Objective 2). It ends with the tests of the Model (Objective 3). Each section gives the results first and then says what they mean in plain words.

The numbers are small and come from one person, so they show whether the method works and where it is weak. They do not show how well it would work for commuters in general. Three measures are used again and again. Average precision tells how well a forecast puts the moments that really became unusual above the ordinary ones, and a higher value is better. Calibration error tells how far the stated chances are from how often they come true. If the Model says 30% many times, about 30% of those moments should turn out unusual, and the calibration error is the average gap, so 0.03 means about 3 points. A lower value is better. A false alarm is an alert that is not followed by a real concern. Alerts that are less than one minute apart are counted as one false alarm episode. Because no trip has a note yet, every trip is counted as an ordinary trip when false alarms are counted.

4.2 The Initial Dataset (Objective 1)

4.2.1 What Was Recorded

The initial dataset is the set of trips of one commuter on one route, in both directions. Table 4.1 describes it. The commuter recorded one trip in each direction on each of 30 recording days, which gives 60 trips and 2,936 records. The trips were recorded in the morning and in the afternoon. None was recorded at night.

Table 4.1. The initial dataset

Item	Value
Commuter	One
Trips	60 (30 from A to B and 30 from B to A)
Records	2,936, one every 10 seconds
Recording period	18 August to 8 October 2026 (52 days)
Recording days	30, with one trip in each direction on each day
Time of day	29 trips in the morning (7:00 to 9:59), 3 at midday, 28 in the afternoon (14:00 to 17:59), none at night
Length of a trip	8 minutes on average (from 5.5 to 11.2 minutes)
Distance of a trip	2.6 km on average (from 2.2 to 3.6 km)
Speed while moving	Middle value 25.5 km/h (most records between 16 and 33 km/h)
Records with the vehicle stopped	24%
Stops	200 stops during the trips, about 3 per trip. The middle length is 30 seconds and the longest is 90 seconds. Each trip also starts with the vehicle standing still for 10 seconds.

4.2.2 Cleaning

No record broke the speed limit of 50 m/s or the acceleration limit, and there were no repeated times or unreadable rows. The file has no accuracy column, so the limit on the error radius was not used. No trip was set aside. The fastest speed in the data was 41 km/h, which is about 11 m/s and far below the 50 m/s limit, so the speed rule had nothing to remove. The cleaning rules therefore removed nothing from this dataset. To check that the rules work, a jump of about 15 km was added to a copy of one trip, and the speed rule removed that one record. This check was made on a copy and is not part of the dataset.

4.2.3 Trips That Leave the Route

The vehicles do not always stay on the route line. Of the 60 trips, 26 stayed within 150 m of the line all the time, and 34 went beyond 150 m at least once. An episode is one continuous stretch of time. An episode away from the route starts when the vehicle goes beyond 150 m of the route line and ends when it is back within 150 m. Together the 34 trips had 41 such episodes.

An episode lasted from 10 to 140 seconds, with a middle value of 60 seconds. The largest distance was 393 m. This is in the Minor band (150 to 500 m), so no trip reached the Major band (500 to 1,000 m) or the Critical band.

Figure 10 shows two trips. Trip 17 stays within the limit. Trip 36 leaves the limit twice and returns each time. These departures happen in more than half of the trips, so for this commuter they are part of normal travel.

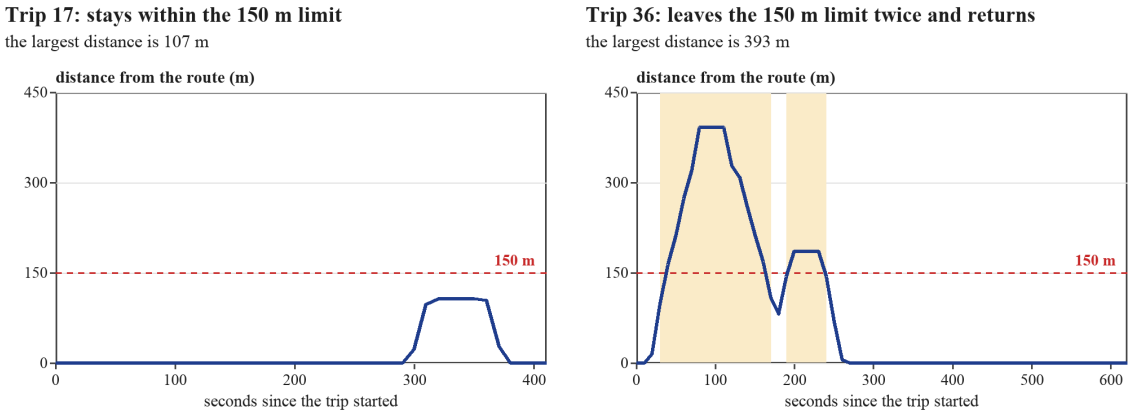


Figure 10. Distance from the route line over time for two trips (the shaded parts are beyond 150 m).

4.2.4 Route Familiarization

A commuter who takes the same detour again and again is doing nothing unusual. The Model handles this in the distance from the route. While the commuter has fewer than 5 earlier trips in a direction, the distance is measured from the route line. After that, it is measured from the commuter's own earlier trips. Figure 11 shows the 30 trips from A to B. Most follow the route line, but several take the same few other paths, in the town near A and around the campus near B.

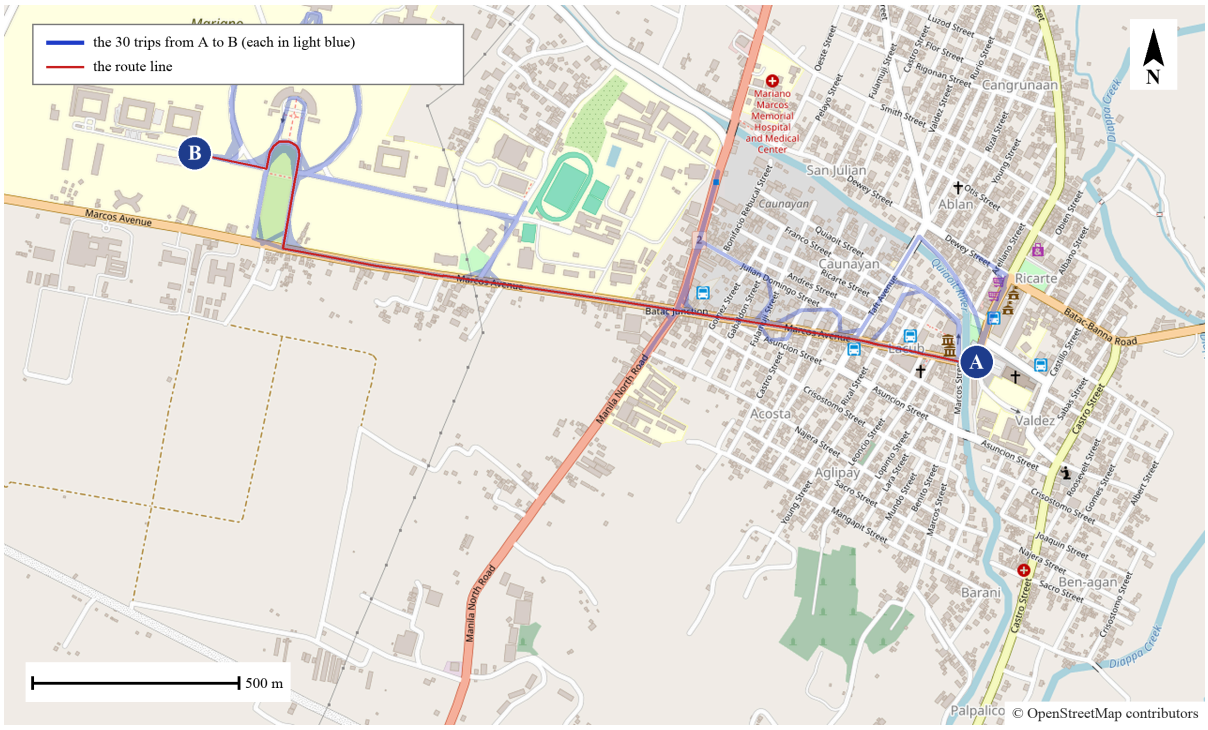


Figure 11. The 30 trips from A to B on the map, with the route line. Map data © OpenStreetMap contributors.

Table 4.2 shows what the earlier trips do. It uses the 11 trips that leave the 150 m limit and have at least 20 earlier trips in the same direction. Measured from the route line, these trips had 106 moments beyond 150 m. With only 2 earlier trips as the history, 56 moments were still beyond 150 m. With 5 earlier trips there were 18, and with 20 earlier trips only 7. This means that 83% of the moments away from the route line were already familiar after 5 earlier trips, and 93% after 20. The commuter's usual detours stop counting as off-route once there is enough history.

Table 4.2. Moments beyond 150 m when the distance is measured from different histories

Distance measured from	Moments beyond 150 m	Share of the moments beyond the route line
The route line	106	100%
2 earlier trips	56	53%
5 earlier trips	18	17%
10 earlier trips	12	11%

20 earlier trips	7	7%

4.3 The Model on the Initial Data (Objective 2)

4.3.1 Concern Levels

The Model gave every moment of every judged trip a concern level. Table 4.3 shows how often each level occurred. Most moments were Normal. The label of a moment is the worst level in the next minute, so labels reach higher levels more often than the levels at the moments themselves.

Table 4.3. Concern levels of the 2,727 judged moments

Level	Moments at that level now	Moments with that level as the label
Normal	2,532 (93%)	2,258 (83%)
Minor	110 (4%)	214 (8%)
Major	41 (2%)	106 (4%)
Critical	44 (2%)	149 (5%)

4.3.2 What Sets the Levels

The level of a moment is the highest of three parts: the Isolation Forest score, the distance from the route, and the length of the current stop. Table 4.4 shows which part reached the level. A moment is counted under more than one column when two parts reach the same level. Route distance reached only Minor, because no trip went beyond 500 m. The Major and Critical levels therefore came from the Isolation Forest score and from long stops, in about equal parts.

Table 4.4. The part that set the concern level

Level now	Moments	Isolation Forest score	Distance from the route	Stop length
Minor	110	41%	38%	25%
Major	41	49%	0%	51%
Critical	44	39%	0%	61%

4.3.3 A Worked Trip

Figure 12 follows Trip 36 from start to end. The Model judged this trip using only the 35 trips recorded before it. The vehicle began to move away from the route at about 20 seconds and passed the 150 m limit at about 40 seconds, when the concern level rose to Minor. When the vehicle stopped at about 90 seconds, the level rose to Major and then to Critical. The level returned to Normal after the vehicle came back to within 150 m of the route.

The forecast did well in one way and badly in two. The chance of Minor or worse rose from 0.02 at 20 seconds to 0.40 at 30 seconds, ten seconds before the level itself rose, and it passed 0.5 at 50 seconds. But the chance of Major or worse stayed at zero during the whole first departure, even when the level was already Critical. Near the end of the trip, the vehicle stopped at about 450 seconds and the chance of Major or worse rose above 0.5. The stop ended before it became long, so Major was never reached and this alert was a false alarm.

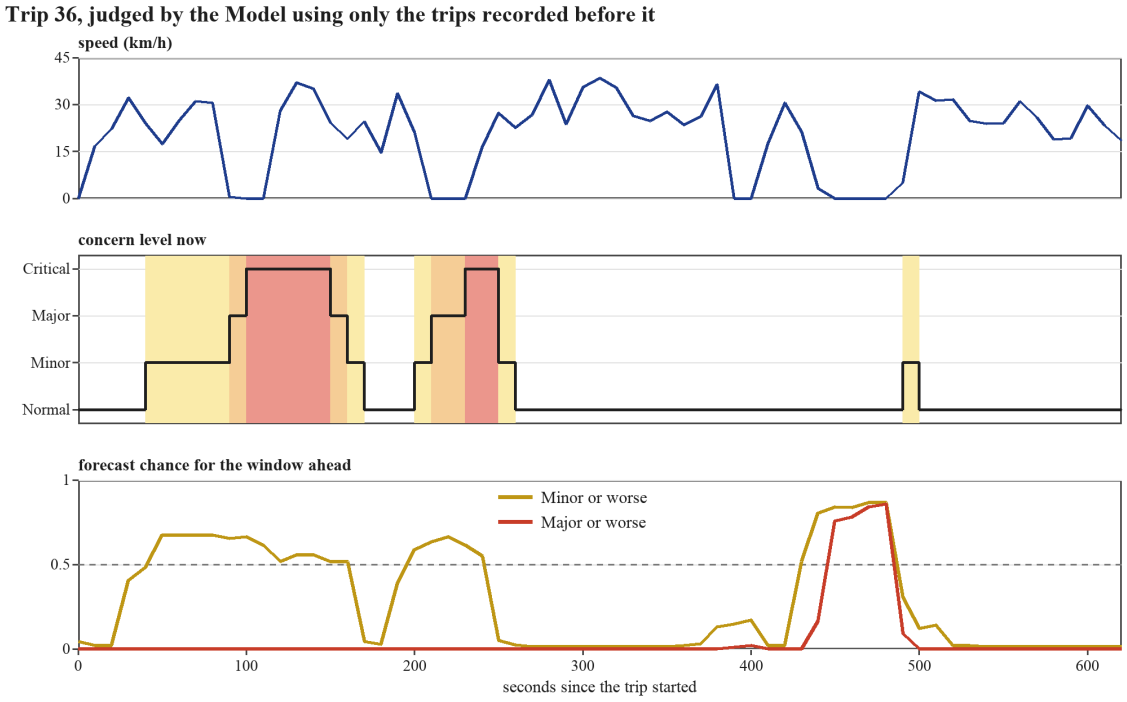


Figure 12. Trip 36: speed, the concern level now, and the forecast chances for the window ahead.

4.4 Evaluation Results (Objective 3)

The trips were judged in date order, each one using only the trips recorded before it. The first five trips were only used as history, so 55 trips and 2,727 moments were judged. Unless a section says otherwise, the Model kept all earlier trips, and the forecast window was one minute.

4.4.1 The Forecast Compared with the Baseline

The baseline assumes that the current concern level continues. Table 4.5 compares it with the forecast. An alert from the forecast means a chance of at least 0.5. An alert from the baseline means that the current level is at or above the level in question.

Table 4.5. The forecast compared with the baseline

Measure	Forecast	Baseline
Average precision, Minor or worse (higher is better)	0.50	0.34
Average precision, Major or worse (higher is better)	0.30	0.20
Minor or worse: share of moments the alert caught	46%	30%
Minor or worse: share of alerts that were right	58%	73%
Major or worse: share of moments the alert caught	19%	21%
Major or worse: share of alerts that were right	41%	62%
Average gap to the true level (lower is better)	0.36	0.28
False alarm episodes per trip	0.58	0.58
False alarm episodes per hour of travel	4.2	4.2

The forecast ranks the unusual moments better than the baseline, for both Minor and Major. For Minor or worse it also catches more moments (46% against 30%), with less accuracy in its alerts. For Major or worse it does not do better than the baseline. It catches about the same share of moments, and fewer of its alerts are right. The average gap to the true level is also larger for the forecast. The two kinds of false alarm are equal at about 0.58 episodes per trip. The trips last about 8 minutes, so this is about 4 false alarms for every hour of travel.

The stated chances can be trusted fairly well. The calibration error was 0.05 for Minor or worse and 0.03 for Major or worse, which means the chances are off by only a few points on

average. The confidence sets are meant to hold the true level 90% of the time. They held it 86% of the time, a little under the target, and each set had 1.2 levels on average.

4.4.2 Warning Ahead of the Current Level

A forecast is useful only if it warns earlier than simply waiting for the current level to rise. Table 4.6 compares the two in the trips where an alert for Major or worse occurred.

Table 4.6. Alerts from the forecast and from the current level (Major or worse)

Result	Trips
Both alerted	18
Only the forecast alerted	11
Only the current level reached Major or worse	11
Where both alerted: the forecast alerted first	78% of the 18 trips
Where both alerted: the middle lead of the forecast	40 seconds

In the 18 trips where both alerted, the forecast came first in 14 of them, with a middle lead of 40 seconds. This is the best result for the forecast. The other rows show its weak side. In 11 trips only the current level reached Major, so the forecast missed them. In 11 trips only the forecast alerted, so those alerts were not followed by a Major level. Over all the judged moments, the forecast alerted in only 15% of the 85 moments where the current level was already Major or worse.

4.4.3 How the Model Behaves as the History Grows

At the start, the Model has few earlier trips to learn from. Table 4.7 groups the judged trips by the number of earlier trips in the history. Figure 13 shows the same results as bars.

Table 4.7. Results by the number of earlier trips in the history

Earlier trips	Trips judged	Average precision, forecast	Average precision, baseline	Calibration error	Confidence set held the true level	False alarms per trip
5–9	5	0.53	0.32	0.08	79%	0.60
10–19	10	0.39	0.26	0.08	82%	1.00

20–29	10	0.37	0.08	0.04	87%	0.50
30–39	10	0.30	0.20	0.06	87%	0.90
40+	20	0.16	0.20	0.02	89%	0.25

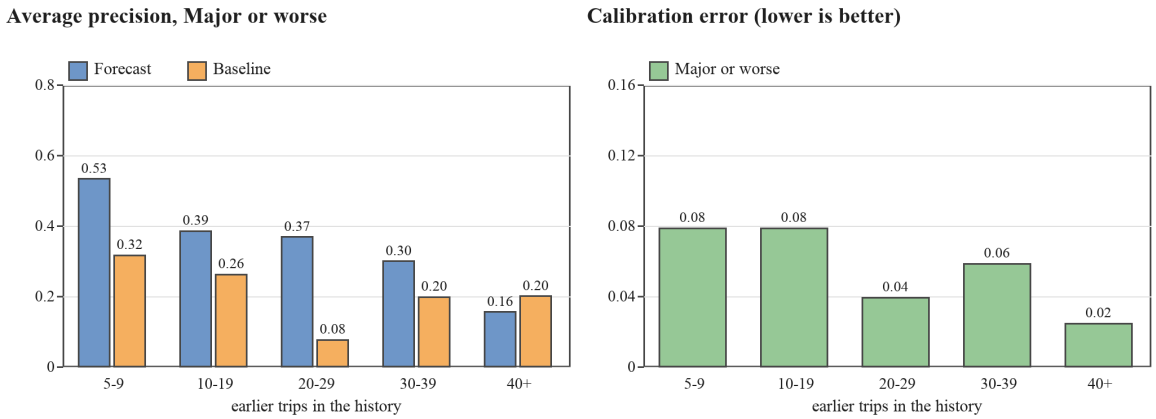


Figure 13. Average precision and calibration error by the number of earlier trips in the history (Major or worse).

The stated chances became more trustworthy as the history grew. The calibration error fell from 0.08 with 5 to 9 earlier trips to 0.02 with 40 or more, and the confidence sets came close to the 90% target in the last group (89%). The forecast ranked better than the baseline in the first four groups. In the last group, with 40 or more earlier trips, the baseline was ahead (0.20 against 0.16). The groups are small, from 5 to 20 trips, so these differences are rough and should not be over-read.

4.4.4 How Long the History Is Kept

The Model can keep all earlier trips or only the recent ones. Table 4.8 compares keeping all of them with keeping only the last 14 days or the last 7 days.

Table 4.8. Results for different history lengths

History kept	Earlier trips used (average)	Average precision, Minor or worse	Average precision, Major or worse	Calibration error, Major or worse	False alarms per trip	False alarms per hour

All earlier trips	32.0	0.50	0.30	0.03	0.58	4.2
Last 14 days	13.8	0.50	0.33	0.04	0.51	3.7
Last 7 days	7.2	0.60	0.37	0.04	0.40	2.9

The results are mixed, and a shorter history is not clearly worse. The false alarms fell as the history got shorter (0.58, 0.51 and 0.40 per trip), and the ranking of Major or worse stayed the same or improved (0.30, 0.33 and 0.37). The trips in this dataset do not change over time, so this dataset cannot show whether old trips become misleading. The choice has to be made with more trips.

4.4.5 Starting Values That Were Tested

Some settings have no direct source, so they were treated as experiments. Table 4.9 shows what happened when each was changed. The "gain" is the forecast's average precision for Major or worse minus the baseline's. A larger gain means that the forecast beats the baseline by more. The last column is the share of moments whose label is Minor or worse.

Table 4.9. Results when the starting values are changed

Setting	Average precision, forecast	Average precision, baseline	Gain	Calibration error	False alarms per hour	Middle lead (seconds)	Labels Minor or worse
Forecast window 30 seconds	0.32	0.24	0.08	0.02	4.2	20	11%
Forecast window 60 seconds (starting value)	0.30	0.20	0.10	0.03	4.2	40	17%
Forecast window 120 seconds	0.29	0.22	0.07	0.06	3.6	70	27%
Road stretch 50 m	0.36	0.24	0.12	0.03	4.2	40	18%
Road stretch 100 m (starting value)	0.30	0.20	0.10	0.03	4.2	40	17%
Road stretch 200 m	0.30	0.21	0.09	0.03	3.8	45	18%
Strength 4 trips	0.33	0.21	0.12	0.04	4.4	40	18%
Strength 8 trips (starting value)	0.30	0.20	0.10	0.03	4.2	40	17%
Strength 16 trips	0.35	0.22	0.13	0.03	4.5	40	17%

Four parts of the day (starting value)	0.30	0.20	0.10	0.03	4.2	40	17%
Day and night only	0.35	0.22	0.13	0.03	4.1	40	17%
No split by time of day	0.35	0.22	0.13	0.03	4.1	40	17%
Earlier trips held out: 10%	0.33	0.20	0.13	0.03	4.0	40	17%
Earlier trips held out: 20% (starting value)	0.30	0.20	0.10	0.03	4.2	40	17%
Earlier trips held out: 30%	0.31	0.20	0.11	0.04	4.8	40	17%
Route distance: the larger of the line and earlier trips	0.30	0.20	0.10	0.04	4.4	40	27%
Route distance: earlier trips after 3 trips	0.29	0.20	0.09	0.04	4.4	40	16%
Route distance: earlier trips after 5 trips (starting value)	0.30	0.20	0.10	0.03	4.2	40	17%
Route distance: earlier trips after 10 trips	0.30	0.20	0.10	0.04	4.2	40	19%

A longer forecast window gives more warning. The middle lead was 20, 40 and 70 seconds for windows of 30, 60 and 120 seconds. The gain stayed between 0.07 and 0.10, but the calibration error was larger for the longest window (0.02, 0.03 and 0.06). The one-minute window is a middle choice between warning time and the trust that can be placed in the chances. The road stretch and the strength changed the results only a little, so the starting values for those two were kept.

The split by time of day did not help. With four parts of the day the gain was 0.10, and with no split it was 0.13. The trips fall in only two parts, morning and afternoon, so the split cut the history of each part in half. The day-and-night version is the same as no split here, because no trip was recorded at night. This dataset therefore cannot show whether time of day matters, and the split is kept as an option until there are trips at other times, especially at night.

The rule for the route distance made a clear difference to the labels. With the earlier rule, which took the larger of the distance from the route line and from earlier trips, 27% of the moments had a label of Minor or worse. With the starting rule it was 17%, because the commuter’s usual detours no longer counted. The results for Major or worse hardly changed

(gain 0.10 and 0.10), and the false alarms stayed close (0.60 and 0.58 per trip). The reason is that Major and Critical never came from route distance in this dataset (Table 4.4).

The share of earlier trips held out for calibration also made a small difference. Holding out 10% instead of 20% raised the average precision for Major or worse from 0.30 to 0.33 and lowered the false alarms from 4.2 to 4.0 per hour. Holding out 30% gave 0.31. The Model has little data, so giving more of it to training helps a little.

The alert limit is a choice too. Table 4.10 shows what happens when the limit on the forecast chance of Major or worse is lowered from 0.5. A lower limit catches more of the Major or worse moments, but it raises the false alarms. At a limit of 0.3, the alerts caught 40% of the moments and 42% of the alerts were right, but there were about 7 false alarm episodes an hour instead of 4. No limit fixes the problem. A limit is a trade between missing real concerns and alerting too often.

Table 4.10. The alert limit for Major or worse

Alert limit (chance of Major or worse)	Share of Major or worse moments caught	Share of alerts that were right	False alarm episodes per hour
0.5 (starting value)	19%	41%	4.2
0.4	31%	43%	6.2
0.3	40%	42%	6.7
0.2	50%	40%	8.6
0.1	60%	32%	9.2

4.4.6 A Fixed Distance Rule for Comparison

A fixed rule is a simple alternative. It raises an alert whenever the vehicle is more than 150 m from the route line, which is the first level of the fixed bands used in SayfTrip (Edwards & Hosein, 2025). Table 4.11 compares it with the forecast of Major or worse over the 55 judged trips. A long stop is a finished stop of 40 seconds or more, the length that Sabir et al. (2025) call brief. A rule counts as flagging a stop or a detour if it alerts from one minute before it starts to its end.

Table 4.11. A fixed distance rule and the forecast of Major or worse

Measure	Fixed 150 m rule	Forecast of Major or worse
Alert episodes per trip	0.69	0.58
Alert episodes per hour of travel	5.0	4.2
Trips with at least one alert	32 of 55	29 of 55
Long stops flagged (56 stops)	6 (11%)	28 (50%)
Episodes beyond 150 m flagged (39 episodes)	39 (100%)	6 (15%)

The number of alerts is about the same, but the alerts are for different things. The fixed rule alerts on every detour, including the routine ones, and it flagged only 6 of the 56 long stops, and only by chance when a stop fell inside a detour. The Model flagged 28 of the long stops and 6 of the 39 detours. Its alerts for detours are fewer because most detours are familiar. There are no trip notes yet, so this does not show which alerts were right. It shows that the Model watches for stops as well as for detours, and that it does not alert on every usual detour.

4.4.7 A Simpler Model for Comparison

XGBoost may be more than this amount of data needs. As a check, the forecast was also made with a plain logistic regression, which is a simple method that uses a few of the same inputs and the same labels. Table 4.12 compares the two on the same trips.

Table 4.12. The Model (XGBoost) and a plain logistic regression

Measure	XGBoost (the Model)	Logistic regression
Average precision, Minor or worse	0.50	0.49
Average precision, Major or worse	0.30	0.34
Minor or worse: share of moments the alert caught	46%	25%
Major or worse: share of moments the alert caught	19%	13%
Major or worse: share of alerts that were right	41%	56%
Alerts when the current level was already Major or worse	15%	25%
Calibration error, Major or worse	0.03	0.03
False alarm episodes per hour	4.2	3.2

Middle lead over the current level (seconds)	40	20

For Major or worse, the simple model ranked the moments better (0.34 against 0.30), had more of its alerts right, and had fewer false alarms. XGBoost was better for Minor or worse, where it caught 46% of the moments against 25%, and it warned earlier (a middle lead of 40 seconds against 20). With only 60 trips, XGBoost has no clear advantage for the rare Major or worse moments. Its advantage is in the more common Minor moments and in earlier warning. These results come from the same trips that were used to compare the settings, so they should be confirmed on new trips.

4.4.8 Results by Direction and Running Time

Table 4.13 shows the results for each direction. They are alike in the number of false alarms. The forecast beat the baseline in both directions.

Table 4.13. Results by direction (Major or worse)

Direction	Trips judged	Average precision, forecast	Average precision, baseline	False alarms per trip	False alarms per hour
A to B	27	0.25	0.17	0.59	4.5
B to A	28	0.37	0.23	0.57	4.0

The Model is quick. Fitting it on the earlier trips took about 0.25 seconds (at most 0.48), and scoring a whole trip took about 0.02 seconds (at most 0.04) on the researcher’s computer. This is small enough to refit the Model after every trip.

4.4.9 Noted Unusual Events

This check could not be made yet, because no trip notes have been collected. The count of noted events flagged by the Model, and the false alarms on trips with no noted event, are left for the next results. The tool that reads the notes is ready. As a check, one stop of about 140 seconds was added to a copy of a trip with a note. The Model flagged it, and the forecast alerted 80 seconds before the noted time. This was only a check of the tool and is not a result.

4.5 Discussion

The first question asked how the trips can be recorded, cleaned, and turned into measures. This worked. The 60 trips had no data errors, and the measures of speed, stopping, and distance from the route could be worked out for every moment. The data showed three things that matter for the Model. The vehicle stopped for about a quarter of the records, it left the route line for a short while in more than half of the trips, and it took the same few other paths again and again. A fixed rule would treat many ordinary trips as unusual.

The second question asked how an Isolation Forest and XGBoost can be used together. The whole pipeline ran from the records to a forecast for every moment. The concern levels were sensible: most moments were Normal, and the higher levels came with unexpected stops and unfamiliar detours. The route distance learned the commuter's usual detours quickly: after 5 earlier trips, 83% of the moments away from the route line were familiar. The forecast of Minor or worse worked well and sometimes warned before the current level rose. The forecast of Major or worse was weak. Its chance often stayed near zero when the current level was already Major or Critical, and it alerted in only 15% of those moments. Major and Critical moments are rare in these data (about 3% of the moments, and 9% of the labels). For a typical trip, XGBoost had only about 70 Major or worse moments to learn from. Changing its settings did not help, and a plain logistic regression ranked Major or worse better (Table 4.12). The weakness therefore comes from the small amount of data and not from a fault in the method.

The third question asked about accuracy, trust, and false alarms. The forecast beat the simple baseline on average precision, but by a modest margin, and it was not better for Major or worse at the 0.5 limit. The stated chances were close to how often they came true, and they improved as the history grew. The confidence sets held the true level 86% of the time, a little under the 90% target. The false alarms are the main problem. About 0.58 alert episodes per trip, or about 4 an hour, would be too many for real use, because people stop answering alerts that are often wrong (Cvach, 2012; Edwards & Hosein, 2025). The comparison with the fixed rule shows

that the Model alerts on different things, not on fewer, and the number of alerts has to come down.

These results do not yet show that the Model is useful in the real world. They show that the method runs quickly, that the forecast can give some warning (a middle lead of 40 seconds where both alerted), that its chances can be trusted fairly well, and that it learns a commuter's usual detours. They also show what must improve: the forecast of Major or worse, and the number of false alarms. Two results were unexpected. The split by time of day did not help, and with 40 or more earlier trips the baseline caught up with the forecast. Both may be caused by the small amount of data, and they should be checked again with more trips.

The trips also had no real incidents, and no notes exist yet. The Model flags what is unusual compared with the commuter's earlier trips, and unusual does not mean unsafe. A long stop or a detour can have an ordinary reason. The notes will show how often the Model's alerts match what the commuter felt was unusual, and that is the check that matters most.

4.6 Limitations and Next Steps

The results come from one commuter on one route, and from 60 trips with no night trips. The groups used for the history curve are small. Many settings were compared on the same trips, so a value that looks best here may be fitted to these trips by chance (Cawley & Talbot, 2010). Any value that is changed should be confirmed on new trips. The check against noted events is still missing.

The next steps are as follows. Collect trip notes with every trip, and add trips from more commuters, routes, vehicles, and times of day. Record at night, so that the split by time of day can be tested. Check again whether the forecast of Major or worse can be improved, and whether the false alarms can be reduced to a level a commuter would accept. Then repeat the tests in this chapter on the larger data.

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